

NELSON, New ZealandWMO: **935460**Lat: **41.3009S**Lon: **173.2160E**Elev: **20**StdP: **14.69**Time Zone: **12.00 (NZL)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.6	34.5	26.5	20.5	40.1	28.9	22.9	39.8	27.8	55.0	24.2	54.7	2.6	200	0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.7	75.9	65.2	73.9	64.7	72.4	64.0	68.3	72.8	67.1	71.5	65.9	70.4	13.5	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	97.7	70.7	65.3	93.4	69.8	64.0	89.2	68.9	32.6	73.1	31.6	71.5	30.7	70.4	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.2	22.8	19.8	DB	29.4	81.1	1.4	2.9	28.3	83.2	27.5	84.9	26.7	86.5	25.6	88.6	(4)
(5)			WB	28.0	70.4	1.5	1.6	26.9	71.6	26.0	72.5	25.2	73.4	24.0	74.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.8	64.6	64.9	62.0	57.2	52.7	47.6	46.3	48.2	51.3	54.7	57.9	62.1
	DBStd	7.41	3.70	3.52	3.51	3.60	4.28	4.27	3.75	3.49	3.52	3.56	3.93	3.64
	HDD50	354	0	0	0	1	22	97	124	76	27	6	1	0
	HDD65	3479	51	40	104	234	382	521	579	521	412	319	214	102
	CDD50	2455	454	418	372	218	105	26	10	20	64	152	239	377
	CDD65	103	40	37	11	0	0	0	0	0	0	0	2	13
	CDH74	186	84	65	10	0	0	0	0	0	0	0	4	23
	CDH80	13	9	2	0	0	0	0	0	0	0	0	0	2
Wind	WSAvg	7.3	8.9	7.8	7.4	6.3	5.9	5.5	5.3	5.7	7.2	8.9	9.4	9.4
Precipitation	PrecAvg	44.6	3.1	2.6	3.2	3.9	3.9	4.6	3.5	4.2	3.8	4.2	3.6	3.9
	PrecMax	63.1	6.4	7.0	7.1	8.9	9.8	8.0	10.3	9.7	7.2	10.4	7.9	15.8
	PrecMin	31.4	0.9	0.2	0.7	0.4	0.4	2.0	0.4	1.5	0.8	0.9	0.6	0.7
	PrecStd	8.3	1.6	1.7	1.7	2.4	2.0	1.7	2.0	1.7	1.7	2.4	1.9	3.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.8	79.3	75.2	70.1	66.1	60.7	58.5	60.7	64.2	68.8	73.1	76.7
		MCWB	64.1	67.5	63.4	60.6	58.1	54.2	52.0	52.1	54.6	56.9	60.2	63.2
	2%	DB	76.0	75.7	72.7	67.7	63.6	58.5	56.8	58.3	61.8	65.5	70.0	73.1
		MCWB	65.3	66.0	63.6	59.9	57.7	53.6	51.5	51.6	54.4	56.7	59.9	63.2
	5%	DB	73.9	73.8	71.1	65.9	61.9	57.1	55.4	56.5	59.9	63.7	67.6	71.1
		MCWB	65.0	65.4	63.0	59.3	56.8	53.2	50.5	51.1	53.8	56.3	58.8	62.7
	10%	DB	72.1	72.3	69.5	64.4	60.3	55.5	53.7	55.1	58.2	61.9	65.6	69.3
		MCWB	64.2	64.5	62.2	58.4	55.9	51.7	49.2	50.5	53.1	55.5	58.2	61.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.1	70.8	67.6	64.6	61.8	57.6	55.8	55.6	58.1	60.9	64.3	67.9
		MCDB	75.7	75.2	72.0	66.6	63.4	58.6	57.0	57.3	60.9	64.5	69.7	72.3
	2%	WB	68.2	68.6	66.3	62.8	60.0	56.0	53.7	53.9	56.6	59.4	62.5	65.9
		MCDB	72.8	72.9	70.2	65.4	61.8	57.3	55.3	55.9	59.1	62.8	67.2	70.2
	5%	WB	67.0	67.3	65.0	61.3	58.2	54.4	51.7	52.6	55.5	58.2	61.0	64.6
		MCDB	71.5	71.9	69.3	64.1	60.7	56.1	54.0	55.1	58.2	61.6	64.9	68.5
	10%	WB	65.7	65.9	63.6	59.9	56.4	52.2	49.8	51.2	54.3	56.9	59.6	63.4
		MCDB	70.2	70.9	68.2	63.2	59.4	54.6	52.7	54.2	57.3	60.5	63.8	67.4
Mean Daily Temperature Range	5% DB	MDBR	13.2	13.7	14.2	14.1	14.0	14.3	14.6	14.0	13.7	13.7	13.9	12.8
		MCDBR	15.4	15.1	15.3	15.2	14.1	13.6	14.9	14.9	15.1	15.6	15.4	14.7
		MCWBR	9.7	9.7	9.6	9.9	9.5	9.6	10.3	10.2	10.4	10.2	10.0	9.6
	5% WB	MCDBR	12.2	12.8	12.8	11.6	11.4	10.2	12.0	12.1	12.4	12.8	12.4	11.6
		MCWBR	8.8	9.2	9.1	9.0	8.9	8.7	9.8	9.4	9.7	9.9	9.5	8.6
Clear-Sky Solar Irradiance	taub	0.337	0.333	0.316	0.306	0.293	0.289	0.284	0.294	0.319	0.324	0.326	0.333	
	taud	2.576	2.600	2.650	2.661	2.658	2.638	2.655	2.649	2.548	2.560	2.569	2.592	
	Ebn at Noon	314	308	300	282	265	255	266	282	293	307	316	317	
	Edh at Noon	33	30	27	23	19	18	19	22	29	31	33	33	
All-Sky Solar Radiation	RadAvg	2093	1872	1456	972	660	503	582	824	1181	1580	1949	2019	
	RadStd	158	163	76	91	38	37	41	55	87	121	167	161	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.58	N/A	N/A	+1.08	N/A	N/A	N/A	-158	+126	N/A
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page